Survival Prediction Across Clinical Time Points: Diagnosis, Sequencing, and Treatment as Alternative Time Zeros

Introduction

In survival analysis, the choice of time zero is important because it defines the start of follow-up and affects the information available for prediction. In oncology studies, several clinical events can serve as time zero, such as diagnosis, genomic sequencing, and treatment initiation. These stages represent different points in the clinical workflow and provide different covariates.

This project studies survival prediction across these clinical time points using the MSK-CHORD cohort. The main goal is to examine whether prediction improves as more information becomes available. To do this, I built three sequential Cox proportional hazards models corresponding to the diagnosis, sequencing, and treatment stages, and then compared their predictive performance and risk stratification ability.

Variables Used and Data Cleaning

The analysis used the MSK-CHORD clinical sequencing cohort, with overall survival in months as the outcome. The diagnosis-stage dataset included patient identifier, survival months, survival status, current age, gender, highest recorded stage, and

cancer type. Observations with missing age were removed before model fitting, leaving 24,947 patients in the diagnosis-stage cohort.

For the sequencing-stage model, I added tumor mutational burden and four mutation indicators: TP53, KRAS, EGFR, and PIK3CA. One sequencing record per patient was retained to create a patient-level genomic dataset. For the treatment-stage model, I selected the first treatment record for each patient and extracted treatment subtype and time to first treatment. After cleaning, the sequencing-stage cohort also included 24,947 patients, while the treatment-stage cohort included 21,470 patients.

Statistical Methods

I used Cox proportional hazards models to study survival prediction at different clinical stages (Cox, 1972). The outcome was overall survival, and death was treated as the event. Three sequential models were fitted.

The first model represented the diagnosis stage and included only baseline clinical variables: age, gender, highest recorded stage, and cancer type.

The second model represented the sequencing stage and added genomic variables to the baseline clinical covariates. These genomic variables included tumor mutational burden and indicators for TP53, KRAS, EGFR, and PIK3CA mutations.

The third model represented the treatment stage and further added treatment subtype and time to first treatment.

Model performance was evaluated using the concordance index, or C-index. The C-index measures how well the model ranks patients by risk, with larger values indicating better discrimination (Harrell, Lee, & Mark, 1996; Uno et al., 2011).

Because the sequencing-stage model had the highest C-index, I used it for downstream risk stratification. I calculated the linear predictor for each patient and split patients into high-risk and low-risk groups using the median predicted risk score. Kaplan–Meier curves were then used to compare survival between these two groups (Kaplan & Meier, 1958). I also summarized the 12-month survival probability and the median overall survival for each risk group.

Results

The first main result was that genomic information improved survival prediction. The diagnosis-stage model had a C-index of 0.664. After genomic variables were added at the sequencing stage, the C-index increased to 0.681. The treatment-stage model had a C-index of 0.671. Therefore, the largest improvement occurred when genomic information was introduced. The diagnosis-stage and sequencing-stage models both used 24,947 patients with 11,294 events, while the treatment-stage model used 21,470 patients with 10,604 events.

Model	N	Events	C-index
Diagnosis	24947	11294	0.664
Sequencing	24947	11294	0.681
Treatment	21470	10604	0.671

Table 1. Comparison of sequential Cox models across clinical stages

The second main result was that the sequencing-stage model provided strong risk stratification. Using the median predicted risk score, patients were divided into 12,473 high-risk patients and 12,474 low-risk patients. The Kaplan–Meier curves showed clear separation between these groups, with a log-rank p-value below 0.0001. The median overall survival was 25.9 months in the high-risk group and 104.6 months in the low-risk group. At 12 months, the survival probability was 0.706 in the high-risk group and 0.896 in the low-risk group. These results show that the sequencing-stage model can separate patients into groups with very different survival outcomes.

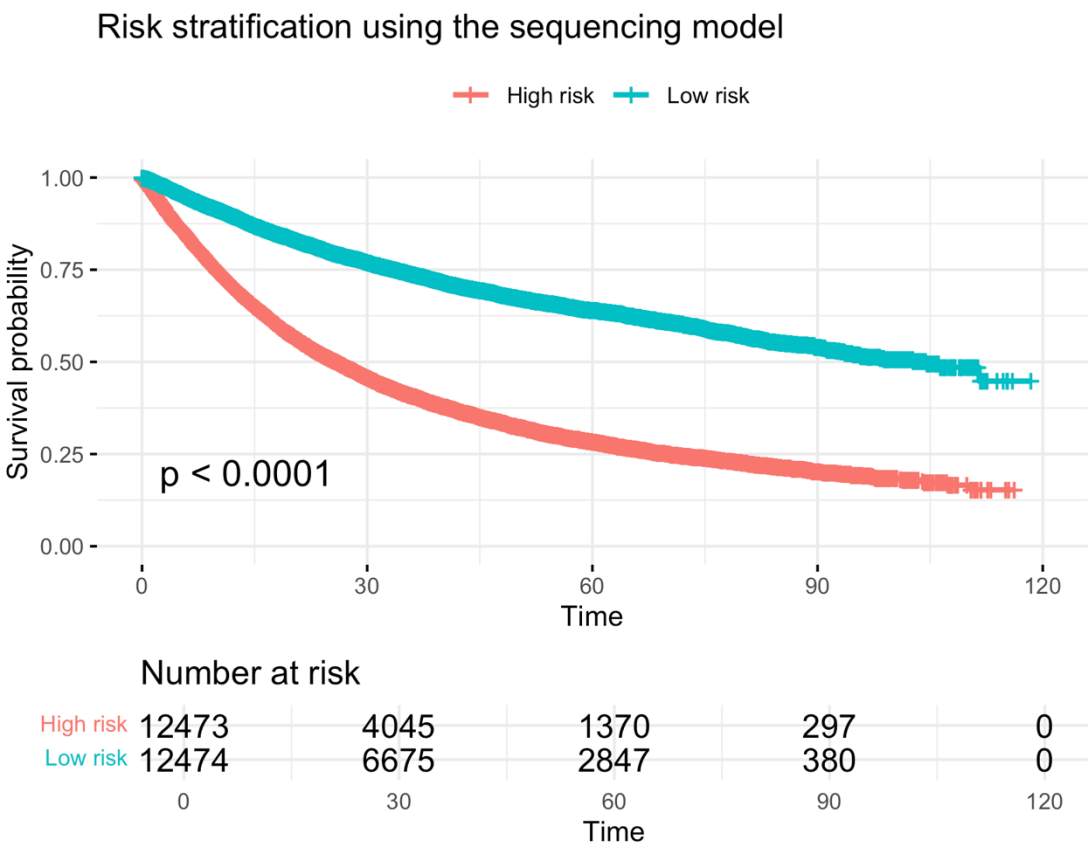

Figure 1. Kaplan–Meier survival curves for high-risk and low-risk groups defined by the sequencing-stage model

The third main result was that several genomic alterations were associated with survival. In the sequencing-stage model, TP53 mutation had a hazard ratio of 1.57, and KRAS mutation had a hazard ratio of 1.32, both indicating higher hazard. EGFR mutation had a hazard ratio of 0.765, indicating lower hazard in this dataset. PIK3CA mutation had a weaker positive association, with a hazard ratio of 1.08. These findings suggest that selected genomic variables carry prognostic information beyond baseline clinical covariates. However, these estimates should be interpreted as associations rather than causal effects.

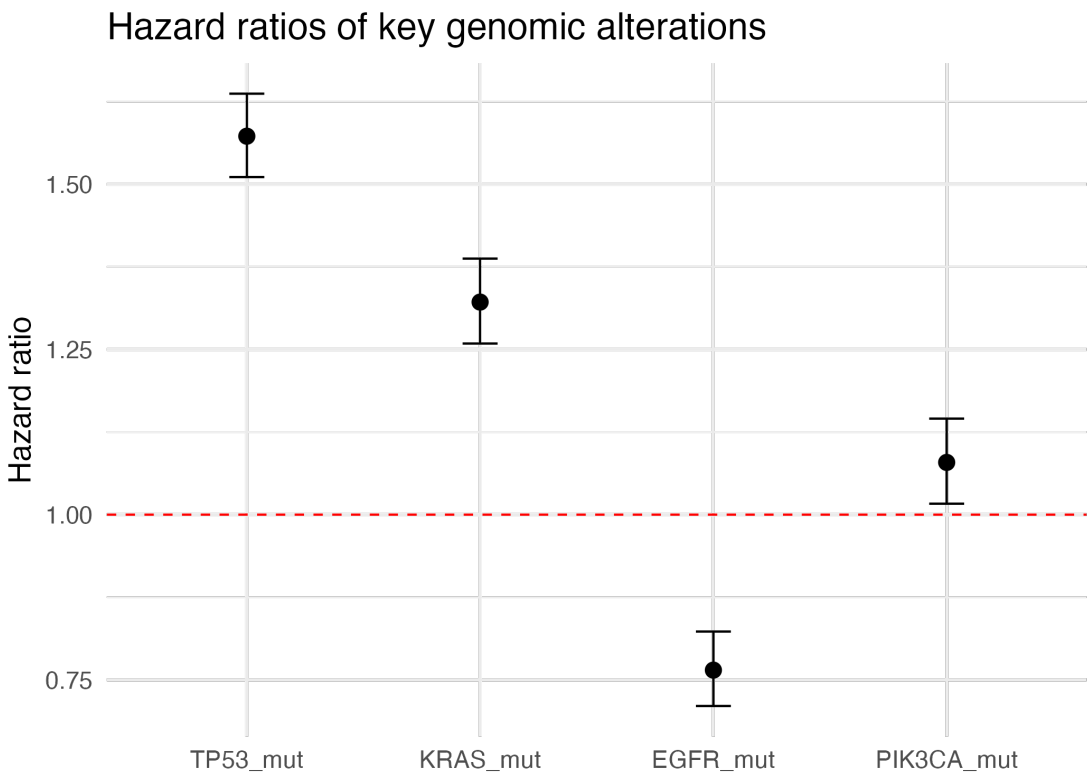

Figure 2. Hazard ratios for selected genomic alterations in the sequencing-stage Cox model

Discussion

This interim analysis showed that genomic information improved survival prediction beyond baseline clinical variables in the MSK-CHORD cohort. Among the three models, the sequencing-stage model performed best and produced the strongest overall discrimination. It also separated patients into high-risk and low-risk groups with large differences in both median overall survival and 12-month survival probability.

The treatment-stage model did not outperform the sequencing-stage model. There are several possible explanations for this. First, treatment information was represented in a simple way, mainly through first treatment subtype and time to first treatment. This may not fully capture the complexity of real treatment history. Second, the treatment-stage cohort was smaller, which may have reduced predictive performance. Third, treatment variables may reflect disease severity and clinical decision making at the same time, making their interpretation more difficult.

This analysis has several limitations. Only four selected genomic alterations were included. The treatment variables were relatively coarse. I also did not fit separate models for individual cancer subtypes. Future work may include adding more genomic features, improving treatment representation, and comparing the Cox model with more flexible machine learning survival methods.

References

Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological), 34*(2), 187–220.

Harrell, F. E., Jr., Lee, K. L., & Mark, D. B. (1996). Multivariable prognostic models: Issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Statistics in Medicine, 15*(4), 361–387.

Kaplan, E. L., & Meier, P. (1958). Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association, 53*(282), 457–481.

Uno, H., Cai, T., Pencina, M. J., D'Agostino, R. B., & Wei, L. J. (2011). On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Statistics in Medicine, 30*(10), 1105–1117.